

Project Overview — Tech Stack & Deployment

A multi-platform social publishing & team collaboration app. Compose a post once and publish or schedule it to YouTube, LinkedIn & X (Instagram coming soon), with team workspaces, a draft → review → approve workflow, analytics, and Pro subscriptions.

Tech Stack

Frontend

- **React + TypeScript**, built with **Vite**
- **Tailwind CSS v4** with a custom light/dark theme
- **React Router** for routing; **React Context** for state

Backend

- **Node.js + Express + TypeScript**
- **PostgreSQL** with **Drizzle ORM** (typed queries + SQL migrations)
- **JWT auth** — short-lived access token + rotating httpOnly refresh cookie
- **bcrypt** password hashing; **AES-256-GCM** for stored OAuth tokens

Integrations

- **OAuth 2.0** to each platform (YouTube / LinkedIn / X), some with PKCE
- **Cloudinary** for media — signed direct-to-browser uploads (large files skip the server)
- **Razorpay** for payments; **Nodemailer** (Gmail SMTP) for password-reset email

React TypeScript Vite Tailwind v4 Node.js Express PostgreSQL Drizzle ORM JWT OAuth 2.0 Cloudinary
Razorpay Nodemailer Docker Nginx PM2 AWS EC2

How It's Deployed to AWS

One **EC2 instance (Ubuntu)**, code pulled from **GitHub**:

- 1 **Nginx** is the front door — serves the built React app and reverse-proxies `/api` to Node on `localhost:4000`.
- 2 **Node API runs under PM2** — auto-restarts and starts on boot via `systemd`.
- 3 **PostgreSQL runs in Docker** (docker-compose) on the same box; its port isn't exposed publicly.
- 4 **HTTPS** via **Let's Encrypt / Certbot** with an auto-renewing certificate.
- 5 Domain from **Hostinger** — an A record points `socialhub.smalltowncoder.in` at the EC2 public IP.
- 6 **Security Group** opens only ports **22 (SSH), 80, 443** — everything else stays closed.

Update flow: SSH in → `git pull` → build server & client → run DB migrations → copy the client build to Nginx's web root → `pm2 restart`.

20-SECOND PITCH

"It's a full-stack TypeScript app — React + Vite on the front, Node/Express + Postgres on the back — that lets you publish to multiple social platforms from one place. I deployed it on a single AWS EC2 box: Nginx serves the site and

proxies the API, the Node server runs on PM2, Postgres runs in Docker, and HTTPS is handled by Let's Encrypt, all behind a Hostinger subdomain."